

Sudhir Kumar Singh

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About me

AI & Data Science student passionate about building production-ready ML systems. I work on projects involving data analysis, machine learning, NLP, GenAI, and full-stack development, turning raw data into useful, real-world applications.

Skills

Languages: C++, C, Java, Python, C#, JavaScript, SQL

Data Analysis: Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn

Machine Learning & AI: Machine learning, NLP, LangChain, LangGraph, Gemini, LLaMA

Frameworks & Tools: FastAPI, Streamlit, MySQL, MongoDB, React.js, Node.js, Express.js, GitHub, Docker

Projects

Stock Price Movement Classification | *Python, Pandas, Scikit-learn, Logistic Regression, Optuna, Streamlit* [Live](#) [GitHub](#)

- Constructed a **time-series-aware ML pipeline** to classify daily stock price movement (**Up/Down**) using 30 years historical NSE: RELIANCE.NS data from Yahoo Finance.
- Evaluated **multiple classifiers** and analyzed performance trade-offs under market efficiency constraints.
- Engineered features with **lag indicators and volume-based metrics**, followed by **hyperparameter tuning** using Optuna.
- Deployed an interactive **Streamlit** dashboard to visualize price trends, model predictions, and evaluation insights.

Predicting Electricity Consumption Based on Weather | *Python, Scikit-learn, Random Forest, XGBoost, Optuna* [Live](#) [GitHub](#)

- Curated a **custom dataset** and built a **machine learning pipeline** to predict daily household electricity consumption.
- Applied **time-series aware preprocessing**, EDA, and feature engineering (lags, rolling averages) to get seasonal patterns.
- Trained and compared **Linear Regression, Random Forest, Gradient Boosting, and XGBoost** on **real-world weather and electricity data**, with **Random Forest** performing best after **Optuna-based tuning**.

AI-Powered YouTube Video Summarizer | *FastAPI, React.js, LangChain, FAISS, Gemini, YouTube Transcript API* [Video](#) [GitHub](#)

- Developed a system to **summarize YouTube videos and answer questions** using Retrieval-Augmented Generation (RAG).
- Integrated **FastAPI backend, React frontend, and a browser extension** for seamless user experience.
- Deployed as a **cross-browser extension**, enabling users to generate summaries and Q&A directly on YouTube platform.
- Implemented **semantic search using FAISS** embeddings to retrieve relevant transcript chunks, improving answer relevance.

Experience

BlueBit 2025 (2nd Place) | *MedGenix: Prescription Scanner & Generic Medicine Recommender* Mar 2025

- Built **MedGenix**, an AI-powered system to scan prescriptions and recommend generic medicines, helping users save up to **85%**.
- Accomplished **92% OCR accuracy** using NLP-based text extraction with fallback logic for handwritten prescriptions.
- Web scraped price data from **10+ pharmacy sources** to enable instant comparison between branded and generic medicines.

HackMatrix 3.0 | *Pregnancy Wellness Platform* Feb 2025

- Created an **AI-enabled platform** for personalized prenatal and postnatal care.
- Designed an appointment alert system with calendar sync, achieving **100% automated reminders**.
- Integrated **meal logging and nutrition analytics**, generating macro-level insights for **3+ meal categories**.

Education

Savitribai Phule Pune University, (B.E) - Artificial Intelligence & Data Science **CGPA: 9.1** **Aug 2023 – June 2027**

Certificates

Machine Learning, DP & NLP	Udemy
The Ultimate Job Ready Data Science Course	CodeWithHarry
Full Stack Web Development Certification	Apna College
C# Programming Certification	Microsoft
Postman API Fundamentals Certification	Postman